

1. Title Page

- **Project Title:** Electric Vehicle Market Trends Analysis
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2. Abstract

- This project is about analysing a comprehensive dataset of 50,000 records detailing global Electric Vehicle (EV) market trends.
- The dataset source includes historical and real-time EV sales, specifications, and regional distribution.
- The problem solved is the lack of integrated visibility into what drives electric vehicle adoption across different demographics.
- The tools used for this analysis include Python, SQL, Excel, and Power BI.
- The final outcome provides stakeholders with a clear view of how battery range, vehicle pricing, and regional infrastructure impact sales, enabling data-driven decisions.

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4. Introduction

- **Why the project is important:** With global EV sales growing exponentially, transitioning from early adopters to mass-market consumers, understanding these shifts is crucial.
- **Industry relevance:** This project evaluates an expansive EV dataset to identify what drives customer purchases, making it highly relevant to automotive manufacturers and infrastructure planners.

- **Use-case of real-time analytics:** Real-time and historical analytics allow businesses to adapt their strategies, optimize pricing, and target regions with the highest growth potential.

5. Problem Statement

- This section clearly defines the issue the project solves.
- Stakeholders do not have real-time visibility into customer search patterns and buying behaviour regarding electric vehicles, making it difficult to identify emerging trends quickly.

6. Objectives

- Analyse real-time search data and sales records.
- Identify rising topics and regional adoption trends.
- Compare keyword popularity and the impact of vehicle price vs. battery range.
- Build visual dashboards for client presentations.
- Provide actionable insights for market expansion.

7. Dataset Description

- **Data source link:** kaggle.
- **File format:** CSV.
- **Number of rows & columns:** 50,001 rows and 13 columns.
- **Variables included:** Date, Country, EV unit sold, EV market share percent, Battery cost per kwh, Charging station, Fast charging, Fuel price, Govt subsidy, Carbon emission , Interest rate, Month EV sales, Sales trend

8. Tools & Technologies Used

Category	Tools
Programming	Python, SQL
Visualization	Power BI, Excel
Data Source	CSV Dataset
Libraries	Pandas, NumPy, Matplotlib

9. Methodology (Workflow)

- Data collection (CSV).

- Data cleaning to handle inconsistencies.
- Exploratory data analysis.
- Visualization of regional and pricing trends.
- Pattern detection for seasonal sales.
- Output generation for the final client report.

10. Data Cleaning

- **Missing values:** Imputed missing battery ranges using the median for the respective vehicle make.
- **Duplicate rows:** Removed duplicate transaction records.
- **Outliers:** Handled extreme price points using statistical methods.
- **Data formatting:** Standardized region names and vehicle models.
- **Date-time conversions:** Converted sales dates into standard formats for time-series analysis.

11. Exploratory Data Analysis (EDA)

- **Line charts:** Showed YoY (Year-over-Year) growth in EV sales.
- **Correlation heatmaps:** Revealed a strong correlation between battery range and vehicle price. * **Peak trend analysis:** Identified Q4 as the peak season for EV purchases.
- **Category performance:** Compared Battery Electric Vehicles (BEVs) against Plug-in Hybrid Electric Vehicles (PHEVs).

12. Insights & Visualizations

- **When demand increases:** Sales consistently spike during the end-of-year holiday season and when new subsidies are announced.
- **Seasonal patterns:** Q4 shows the highest sales volume globally.
- **Regional differences:** Urban regions show a significantly higher adoption rate compared to rural areas.
- **Audience behaviour:** Consumers are increasingly prioritizing battery range over brand loyalty.

13. Results / Output

- **Dashboards:** An interactive dashboard tracking global sales and pricing. * **Trend insights:** Identified a major increase in affordable, mid-range EVs.
- **Alerts or signals:** Configured alerts for regions showing a sudden drop in sales.
- **Example Output:** EV sales increased 45% compared to the previous year, with the highest interest occurring in Q4.

14. Challenges

- **Data irregularities:** Inconsistent naming conventions for vehicle models across different regions.
- **Missing timestamps:** Some records lacked exact sales dates.
- **Data noise:** Fluctuations caused by supply chain disruptions.

- **Complex pattern detection:** Separating the impact of government subsidies from organic consumer demand.

15. Recommendations

- **Run analysis weekly:** To track real-time changes in consumer demand.
- **Expand to multiple regions:** Focus marketing efforts on emerging markets.
- **Include competitor keywords:** Monitor competitor pricing and feature rollouts.

16. Conclusion

- **What was learned:** Battery range and pricing are the most critical factors driving EV adoption.
- **How results help decision-making:** The insights allow the client to optimize inventory and target marketing campaigns effectively.
- **Importance of real-time analytics:** Continuous monitoring is essential in the fast-paced EV market to stay ahead of competitors.

17. Future Scope

- **Add forecasting model:** Implement a machine learning model to predict next quarter's sales.
- **Automate dashboard:** Connect the dashboard directly to a live database for automatic updates.
- **Integrate multiple datasets:** Include public charging station data to map infrastructure against sales